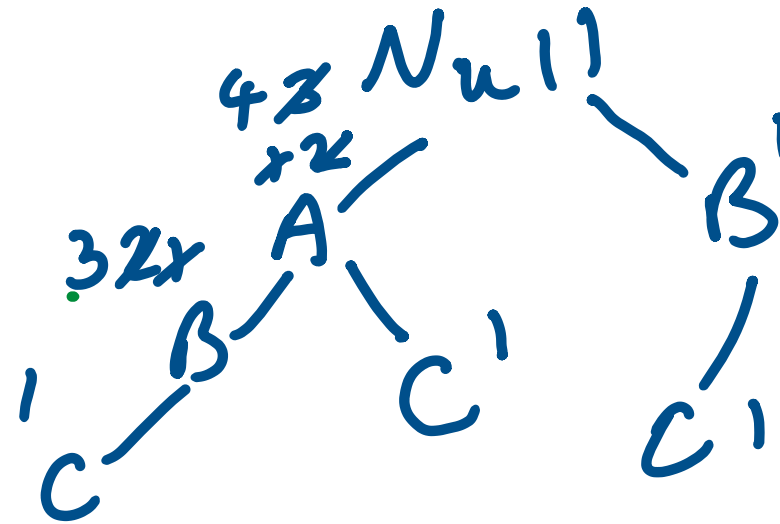


FP Growth

$$L_1 = \{ \bar{A}_4, \bar{B}_4, \bar{C}_3 \}$$

<u>A, B, C</u>	✓	✓
<u>A, C</u>	✓	✓
<u>A, B</u>	✓	✓
<u>A, B</u>	✓	✓
<u>B, C</u>	✓	



FP tree

item

pattern base

$\{A, B\} : 1$

$\{A\} : 1$

$\{B\} : 1$

$\{A\} : 3$

$\{\} : 1$

$\{\} : 4$

C

B

A

item	pattern base			Conditional FP _{tree}
C	$\{A, B\}:1$	$\{A\}:1$	$\{B\}:1$	$\{A\}:2$ $\{B\}:2$
B	$\{A\}:3$	$\{\}:1$		$\{A\}:3$
A	$\{\}:4$			α

item	pattern base	Conditional FP _{tree}
C	$\{A, B\}:1$ $\{A\}:1$ $\{B\}:1$	$\{A\}:2$ $\{B\}:2$ $\{A, C\}$ $\{B, C\}$
B	$\{A\}:3$ $\{\}:1$	$\{A\}:3$ $\{A, B\}$
A	$\{\}:4$	α

minSupport = 2

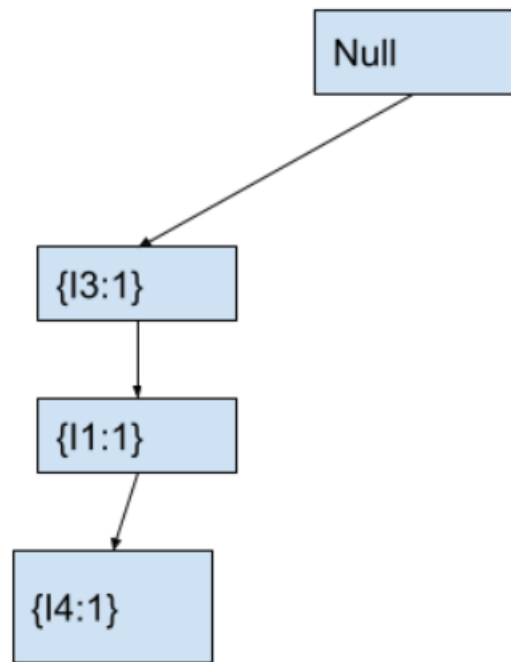
Transaction ID	Items
T1	I1, I3, I4 .
T2	I2, I3, I5, I6
T3	I1, I2, I3, I5
T4	I2, I5
T5	I1, I3, I5

Item	Support Count
I1	3
I2	3
I3	4
I4	1
I5	4
I6	1

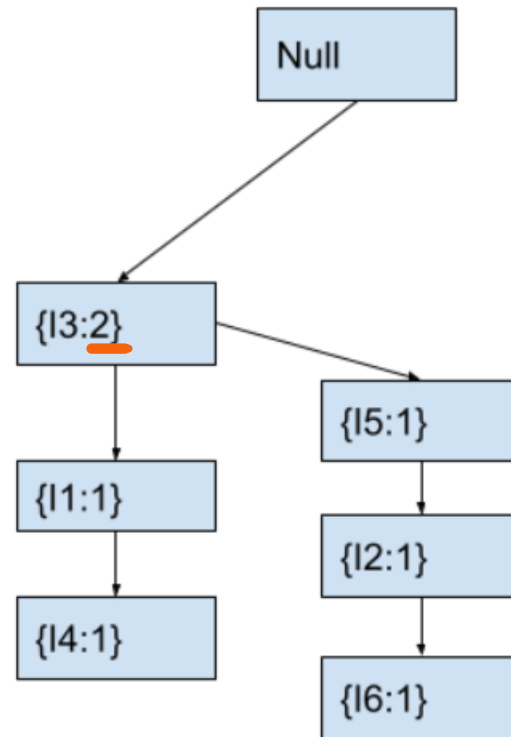
$L_1 = \{I3, I5, I1, I2\}$

Reorganize The Items in The Transaction Dataset

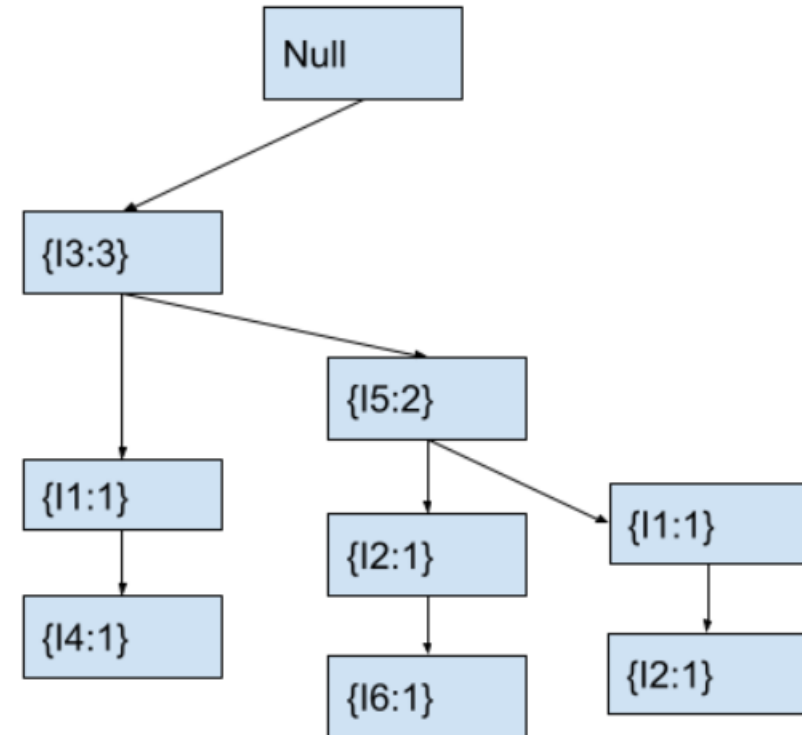
Transaction ID	Items
T1	I3, I1, I4
T2	I3, I5, I2, I6
T3	I3, I5, I1, I2
T4	I5, I2
T5	I3, I5, I1



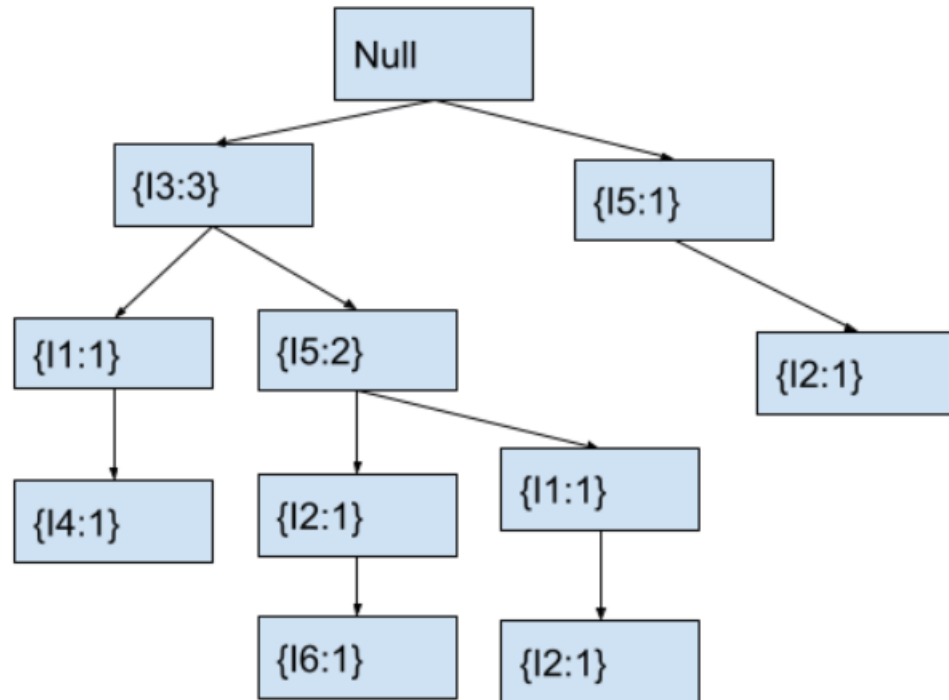
Fp-tree with one transaction



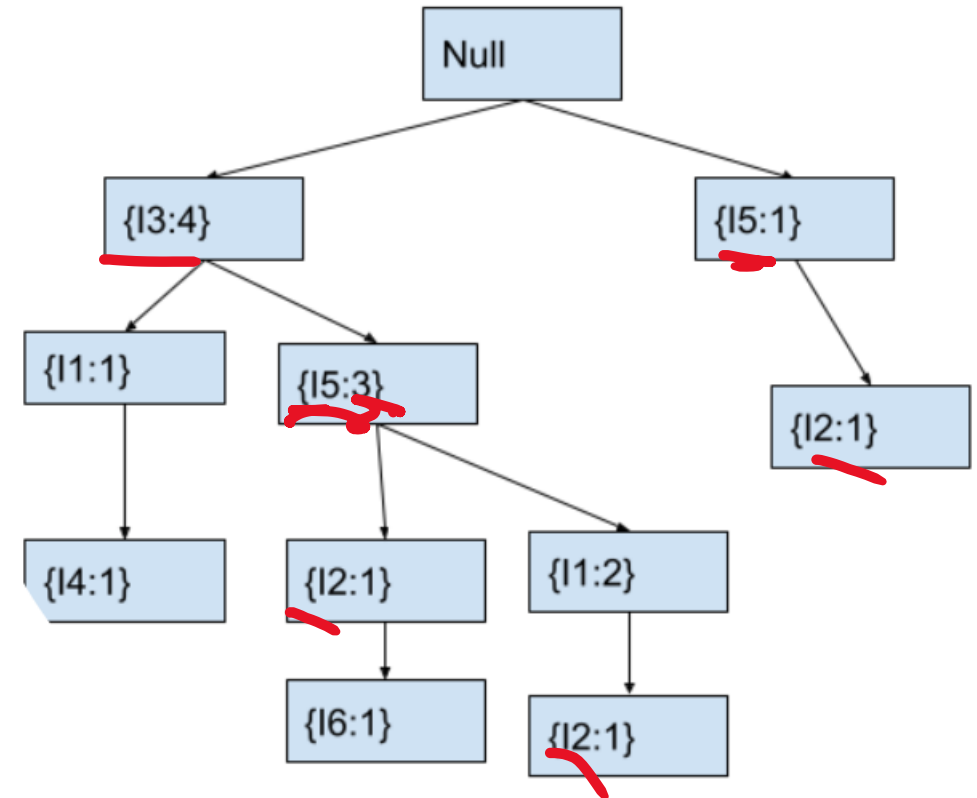
Fp-tree with two transactions



Fp-tree with three transactions

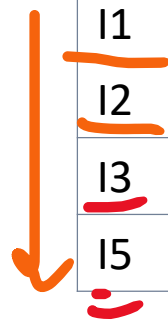


Fp-tree with four transactions



Final FP-Tree for the FP Algorithm Numerical Example

Pattern Base created in FP Growth Numerical Example



Item	Pattern Base
I1	{I3}:1,{I3, I5}:2
I2	{I3, I5}:1,{I3, I5, I1}:1,{I5}:1
I3	{}
I5	{I3}:3

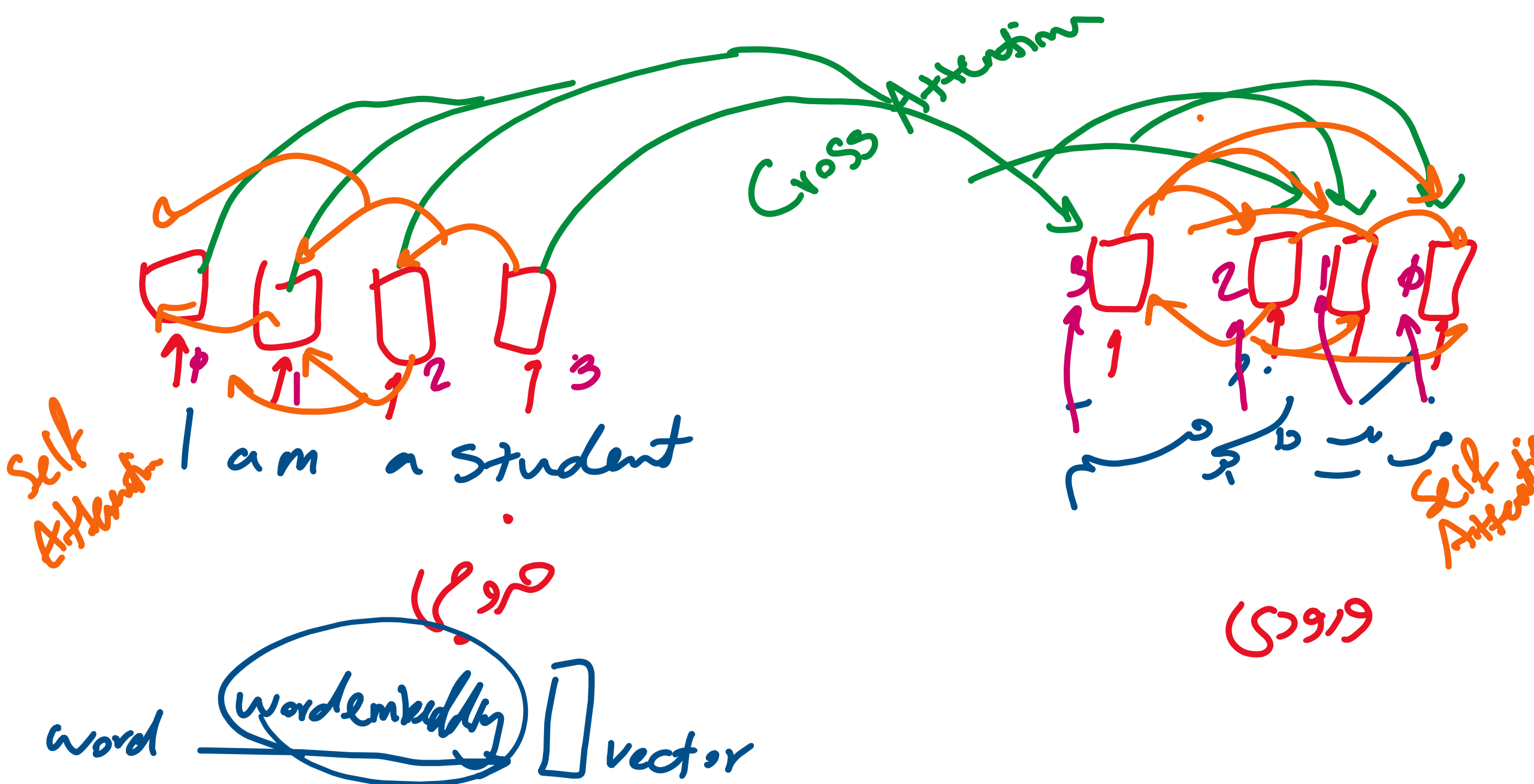
missupp = 2

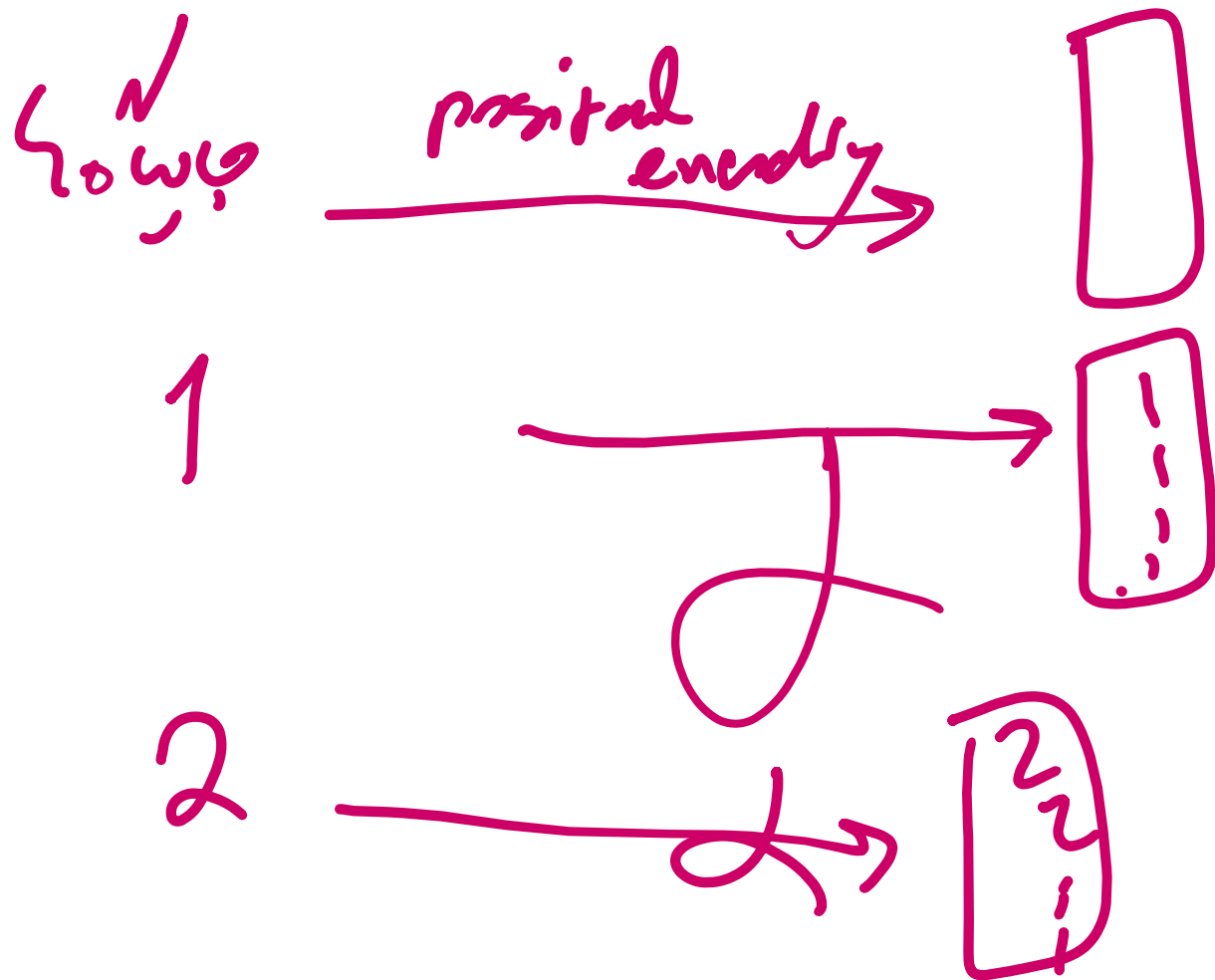
Conditional FP-Tree for items

Item	Pattern Base	Conditional FP Tree
I1	<u>{I3}</u> :1, <u>{I3, I5}</u> :2	<u>{I3:3, I5:2}</u>
I2	<u>{I3, I5}</u> :1, <u>{I3, I5, I1}</u> :1, <u>{I5}</u> :1	<u>{I3:2, I5:3}</u> DA
I3	{}	
I5	{I3}:3	<u>{I3:3}</u>

Frequent itemsets generated in the FP Growth Numerical example

Item	Pattern Base	Conditional FP Tree	Frequent itemsets
I1	{I3}:1, {I3, I5}:2	{I3:3, I5:2}	{I1, I3}:3, {I1, I5}:2, {I1, I3, I5}:2
I2	{I3, I5}:1, {I3, I5, I1}:1, {I5}:1	{I3:2, I5:3}	{I2, I3}:2, {I2, I5}:3, {I2, I3, I5}:2
I5	{I3}:3	{I3:3}	{I3, I5}:3





برای حرف می

فرد پیر

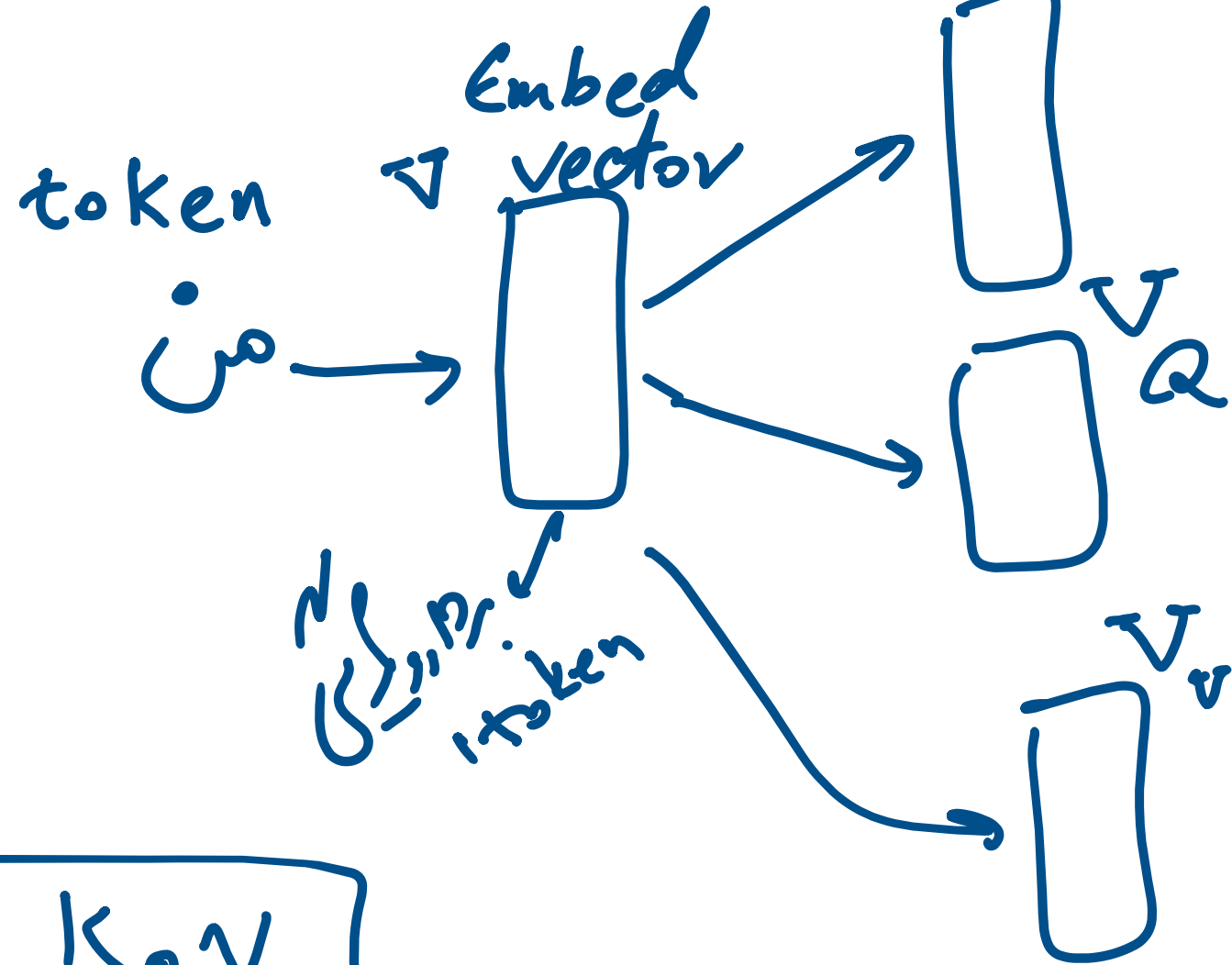
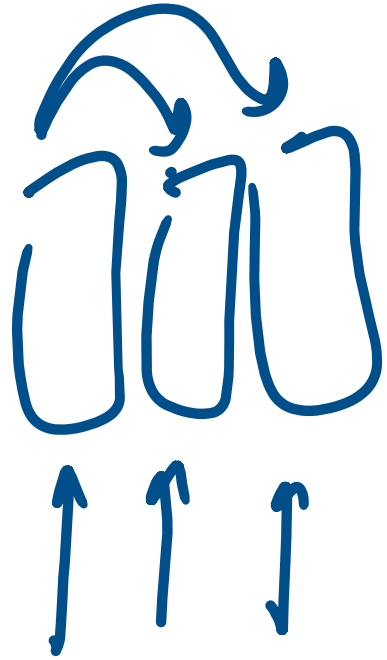
Model

floating point operations

FLOPS ← CPU
GPU

کتابخانه
floating point operations per second flops

tops
int

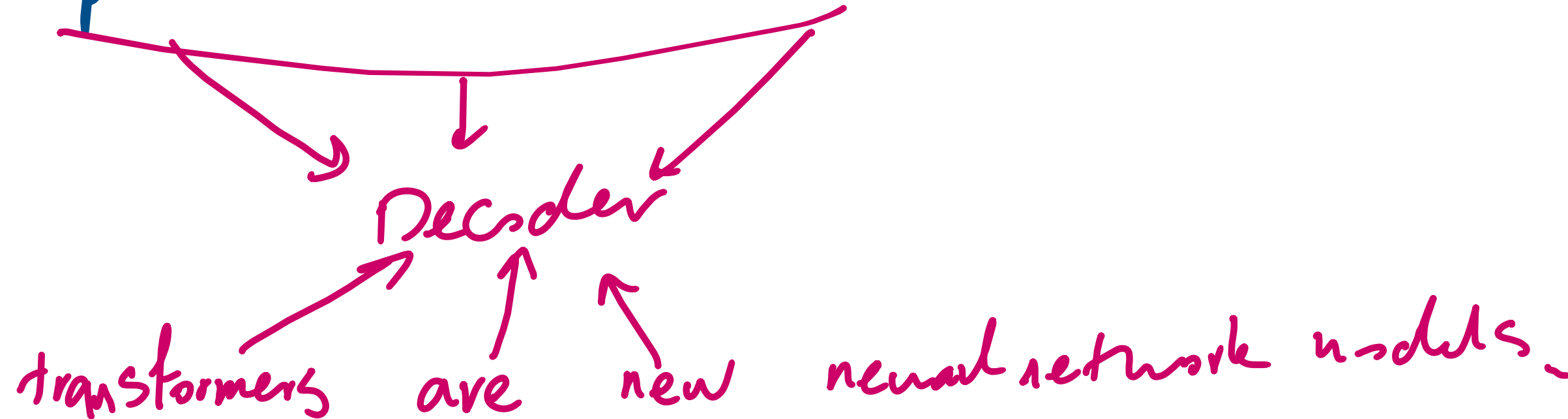


Q, K, V

$$\cos(\vec{Q}, \vec{K}) = \frac{Q \cdot K}{\|Q\| \|K\|}$$

Explain transformer models.

GPT



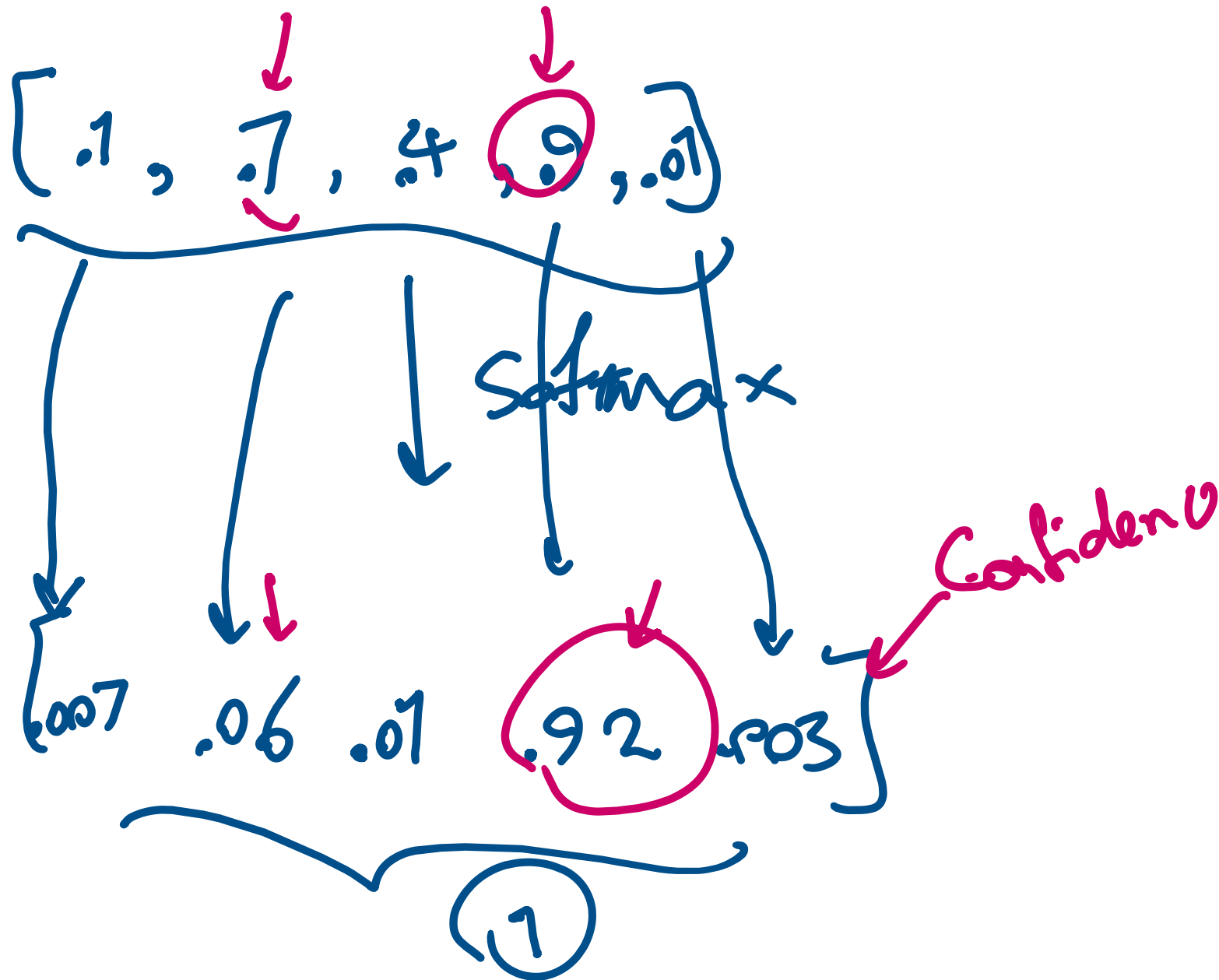
Bigram
 رگبار بار
 (x, y)
 (x, y)
 $P(y|x)$

tri-gram
 رگبار بار
 (x, y, z)
 $P(z|x, y)$

n-gram
 رگبار بار
 $P(u_i | u_{i-n}, \dots, u_{i-1})$

$$\underline{\text{Softmax}(y_1, y_2, \dots, y_n)} \Rightarrow$$

$$\text{Softmax}(y_i) = \frac{e^{y_i}}{\sum_{j=1}^n e^{y_j}}$$



$[.1, .1, .12] \rightarrow [37, 37, .56]$

I am a student .56 ✓ .8 ✓
 teacher .4 .1
 blackboard :
 TV :

